Lecture 22: Vectors

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Based on Thomas' Calculus, 15th Edition in SI Units, Pearson Education Limited

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Learning Outcome

At the end of this lecture, you should be able to:

- ☐ Define a vector in 2D and 3D, including its magnitude, direction, and standard position.
- ☐ Determine the component form of a vector given initial and terminal points.
- ☐ Calculate the magnitude of a vector
- ☐ Perform vector operations and apply vector operations properties.
- ☐ Express a vector as a linear combination of the standard unit vectors.
- ☐ Find a unit vector in the direction of a given vector.
- ☐ Compute the midpoint of a line segment between two points in space.

Introduction

There are two types of quantities we measure

- scalar
- vector

Vectors

A vector quantity is anything that is specified by both a magnitude and a direction.

The vector is represented by a directed line segment.

The directed line segment $\overrightarrow{AB}$ has initial point A and terminal point B and its length is denoted by $\left|\overrightarrow{AB}\right|$.

Two vectors are equal if they have the same length and direction.

Notation: $\boldsymbol{v}$, $\vec{v}$, $\overrightarrow{PQ}$

Standard position vector is the line segment with the initial point at the origin.

Vectors

If $\boldsymbol{v}$ is a two-dimensional vector in the plane equal to the vector with initial point at the origin and terminal point (v_1, v_2), then the component form of $\boldsymbol{v}$ is

$$\boldsymbol{v} = \langle v_1, v_2 \rangle.$$

If $\boldsymbol{v}$ is a three-dimensional vector equal to the vector with initial point at the origin and terminal point (v_1, v_2, v_3), then the component form of $\boldsymbol{v}$ is

$$\boldsymbol{v} = \langle v_1, v_2, v_3 \rangle.$$

If the vector $\boldsymbol{v} = \langle v_1, v_2, v_3 \rangle$ is represented by the directed line segment $\overrightarrow{PQ}$, where $P(x_1, y_1, z_1)$ is the initial point and $Q(x_2, y_2, z_2)$ is the terminal point, then

$$\langle v_1, v_2, v_3 \rangle = \langle x_2 - x_1, y_2 - y_1, z_2 - z_1 \rangle.$$

Length

The magnitude (or length) of the vector
$\boldsymbol{v} = \langle v_1, v_2, v_3 \rangle = \langle x_2 - x_1, y_2 - y_1, z_2 - z_1 \rangle$ is the nonnegative number

$$|\boldsymbol{v}| = \sqrt{{v_1}^2 + {v_2}^2 + {v_3}^2} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}.$$

Example: Find the component form and length of the vector with initial point $P(-3, 4, 1)$ and terminal point $Q(-5, 2, 2)$.

Vector Algebra Operations

Let $\boldsymbol{u} = \langle u_1, u_2, u_3 \rangle$ and $\boldsymbol{v} = \langle v_1, v_2, v_3 \rangle$ be vectors with k a scalar.

Addition: $\boldsymbol{u} + \boldsymbol{v} = \langle u_1 + v_1, u_2 + v_2, u_3 + v_3 \rangle$

Scalar multiplication: $k\boldsymbol{u} = \langle ku_1, ku_2, ku_3 \rangle$

Difference: $\boldsymbol{u} - \boldsymbol{v} = \boldsymbol{u} + (-\boldsymbol{v}) = \langle u_1 - v_1, u_2 - v_2, u_3 - v_3 \rangle$

Vector Algebra Operations

Example: Let $\boldsymbol{u} = \langle -1, 3, 1 \rangle$ and $\boldsymbol{v} = \langle 4, 7, 0 \rangle$. Find the components of

(a) $2\boldsymbol{u} + 3\boldsymbol{v}$

(b) $\boldsymbol{u} - \boldsymbol{v}$

(c) $\left| \frac{1}{2}\boldsymbol{u} \right|$

Properties of Vector Operations

Let $\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}$ be vectors and a, b be scalars.

1. $\boldsymbol{u} + \boldsymbol{v} = \boldsymbol{v} + \boldsymbol{u}$
2. $(\boldsymbol{u} + \boldsymbol{v}) + \boldsymbol{w} = \boldsymbol{u} + (\boldsymbol{v} + \boldsymbol{w})$
3. $\boldsymbol{u} + \boldsymbol{0} = \boldsymbol{u}$
4. $\boldsymbol{u} + (-\boldsymbol{u}) = \boldsymbol{0}$
5. $0\boldsymbol{u} = \boldsymbol{0}$
6. $1\boldsymbol{u} = \boldsymbol{u}$
7. $a(b\boldsymbol{u}) = (ab)\boldsymbol{u}$
8. $a(\boldsymbol{u} + \boldsymbol{v}) = a\boldsymbol{u} + a\boldsymbol{v}$
9. $(a + b)\boldsymbol{u} = a\boldsymbol{u} + b\boldsymbol{u}$

If three or more space vectors lie in the same plane, we say they are coplanar vectors

Unit Vector

A vector $\boldsymbol{v}$ of length 1 is called a unit vector.

The standard unit vectors are

$$\boldsymbol{i} = \langle 1, 0, 0 \rangle, \quad \boldsymbol{j} = \langle 0, 1, 0 \rangle, \quad \boldsymbol{k} = \langle 0, 0, 1 \rangle.$$

Any vector $\boldsymbol{v} = \langle v_1, v_2, v_3 \rangle$ can be written as a linear combination of the standard unit vectors.

Unit Vector

If $\boldsymbol{v} \neq \mathbf{0}$, then

1. $\frac{\boldsymbol{v}}{|\boldsymbol{v}|}$ is a unit vector called the direction of $\boldsymbol{v}$;
2. the equation $\boldsymbol{v} = |\boldsymbol{v}|\frac{\boldsymbol{v}}{|\boldsymbol{v}|}$ expresses $\boldsymbol{v}$ as its length times its direction.

Example: Find a unit vector $\boldsymbol{u}$ in the direction of the vector from $P_1(1, 0, 1)$ to $P_2(3, 2, 0)$.

Unit Vector

Example: Express $\boldsymbol{v} = 3\boldsymbol{i} - 4\boldsymbol{j}$ as a product of its magnitude times its direction.

Unit Vector

Example: Find a vector of magnitude 6, in the direction of the vector $\boldsymbol{v} = 2\boldsymbol{i} + 2\boldsymbol{j} - \boldsymbol{k}$.

Midpoint of a Line Segment

The midpoint M of the line segment joining points $P_1(x_1, y_1, z_1)$ and $P_2(x_2, y_2, z_2)$ is the point

$$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}, \frac{z_1 + z_2}{2} \right).$$

Midpoint of a Line Segment

Example: Find the midpoint of the segment joining $P_1(3, -2, 0)$ and $P_2(7, 4, 4)$.